Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/05/2018
PI Name:	Gur, Ruben C. Ph.D.		
Project Title:	Cognition Longitudinal Validation Study in High-Performing Healthy Controls at Penn		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gur@upenn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-615-3604
Organization Name:	University of Pennsylvania Health System		
PI Address 1:	Hospital of the University of Pennsylvania		
PI Address 2:	Neuropsychiatry Section		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-4283	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/01/2016	**End Date:**	04/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-NBPF00013		
Performance Goal No.:			
Performance Goal Text:			

This is a continuation of Project NBPF00009, Cognition Validation Study in High-Performing Healthy Controls at Penn. Under National Space Biomedical Research Institute (NSBRI) funding we studied 96 high functioning (MS or higher in STEM) individuals (48 males/48 females) with the average female age being 40.35 and the average male age being 40.27. This group was precisely balanced in four age bins (25-32, 33-40, 41-48, and 49-59) by sex. All participants received Cognition (Basner et al., 2016) in both laptop and iPad versions and WinSAT on the laptop, in a counterbalanced order (Latin square design). The results were reported in the final progress report, presented at the Human Research Program (HRP) Investigator Workshop, and are now in press in Aerospace Medicine and Human Performance (Moore et al., in press. [Ed. note Feb 2018: article was in press at time of report submission to NSBRI; now published]). They provide astronaut-appropriate normative data on Cognition.
Since Cognition has been administered in spaceflight and analog environments in schedules ranging from multiple times per day to monthly, we have data on practice effects over short inter-test intervals. However, to transition Cognition into

Task Description:	operations information is needed on performance retention and potential practice effects when the interval is longer without any practice in between. This longitudinal follow-up aims to determine the practice effects on each performance test in the Cognition battery in this sample. The 96 high-performing subjects from the original validation study were re-contacted to participate in follow-up procedures. Additional recruitment efforts were made to replace participants who were unable or unwilling to return. At follow-up, subjects were administered WinSCAT and the same laptop Cognition battery that they had previously received. Half of the subjects took the practice version of each Cognition test before taking the full version again. This will establish whether a practice version of each test is enough to boost performance after a longer time interval without intervening tests. Cognition contains 10 individual tests. The battery evaluates several domains including sensory-motor speed, processing speed, memory, visual abstraction, spatial orientation, emotion recognition, attention, and risk-related decision-making. Reliability and validation are assessed using Classical Test Theory and Item Response Theory. As in the intake study completed last year, participants will also be administered the WinSCAT on the laptop computer for comparisons to the Cognition data. By bringing this sample back, we will evaluate Cognition and WinSCAT performance after a longer time interval without having performed a test during that interval. Since initiating the project at the beginning of this year, 82 individuals have been enrolled and 79 completed both sessions (37 males/42 females) with the average female age being 42.21 and the average male age being 41.11 at time of follow-up. We reached about 84% of the stipulated goal of enrolling 96 individuals in four age bins (25-32, 33-40, 41-48, and 49-59) balanced by sex. The completion rate for all three neurocognitive batteries was 100%. In all, 163 bouts were successfully administered. Though due to the delayed start recruitment goals have not yet been met, we plan to continue enrolling individuals to balance the data-set until goals are met. Of note, we are starting a new NASA-funded imaging protocol, Functional neuroimaging of Cognition challenges in high performing individuals. This study too will draw from the existing pool of high-performing adults. References Basner M, Savitt A, Moore TM, Port AM, McGuire S, Ecker AJ, Nasrini J, Mollicone DJ, Mott CM, McCann T, Dinges DF, Gur RC. Development and Validation of the Cognition Test Battery for Spaceflight. Aerosp Med Hum Perform. 2015, 86(11):942-952. Moore TM, Basner M, Nasrini J, Hermosillo E, Kabadi S, Roalf DR, McGuire S, Ecker AJ, Ruparel K, Port AM, Jackson CT, Dinges DF, Gur RC. Validation of the Cognition Test Battery for Spaceflight in a Sample of Highly Educated Adults. Aerosp Med Hum Perform. (in Press) [Ed. note Feb 2018: article was in press at time of report submission to NSBRI; now published. See Bibliography section]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will provide further longitudinal validation of a dynamic and unobtrusive test battery designed to assess cognitive performance of high-performing adults. Cognition was developed to be used in space by astronauts; however, there are Earth-based benefits that could be provided by the battery. The Cognition battery, once validated longitudinally through this project, would provide assessments of cognitive functioning in high-performing adults who carry out high stress jobs over prolonged periods of time. It would offer milestone performance scores in areas such as decision making, mental abstraction, social cognition, sensory-motor skills, and working memory that would be necessary in the field.
Task Progress:	Since initiating the project at the beginning of 2017, 82 individuals have been enrolled and 79 completed both sessions (37 males/42 females) with the average female age being 42.21 and the average male age being 41.11 at time of follow-up. We reached about 84% of the stipulated goal of enrolling 96 individuals in four age bins (25-32, 33-40, 41-48, and 49-59) balanced by sex. The completion rate for all three neurocognitive batteries was 100%. In all, 163 bouts were successfully administered. Though due to the delayed start recruitment goals have not yet been met, we plan to continue enrolling individuals to balance the data-set until goals are met.
Bibliography Type:	Description: (Last Updated: 03/13/2018)
Articles in Peer-reviewed Journals	Moore TM, Basner M, Nasrini J, Hermosillo E, Kabadi S, Roalf DR, McGuire S, Ecker AJ, Ruparel K, Port AM, Jackson CT, Dinges DF, Gur RC. "Validation of the Cognition Test Battery for spaceflight in a sample of highly educated adults." Aerosp Med Hum Perform. 2017 Oct;88(10):937-46. https://doi.org/10.3357/AMHP.4801.2017 ; PubMed PMID: 28923143 , Oct-2017
Awards	Gur R. "Elected Fellow, American Association for the Advancement of Science (AAAS), November 2016." Nov-2016